Date: __/__/____

Euler's theorem on homogeneous functions

If u be a homogeneous function of degree n in x and y, then

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu.$$

Since, u is a function of degree n in x and y, therefore,

$$\boxed{u = x^n f(y/x)} \rightarrow \text{Condition for homogeneous}$$

$$\therefore \frac{\partial u}{\partial x} = nx^{n-1} f(y/x) + x^n f'\left(\frac{y}{x}\right) \cdot y\left(-\frac{1}{x^2}\right)$$

and $$\frac{\partial u}{\partial y} = x^n f'(y/x) \cdot \frac{1}{x} = x^{n-1} f'\left(\frac{y}{x}\right).$$

Hence, $$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nx^n f(y/x) = nu$$

In general,

if u be the homogeneous of degree n in $x, y, z, t, \cdots$ then,

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} + t\frac{\partial u}{\partial t} + \cdots = nu.$$

Akash

Date: __/__/____

Q.1. Show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 2u\log u$ where, $\log u = (x^3+y^3)/3x+4y$.

Sol: Given;

$$\log u = (x^3+y^3)/3x+4y$$

let $z = \log u = \dfrac{x^3\left[1+\left(\frac{y}{x}\right)^3\right]}{x\left[3+4y/x\right]}$

$$z = \log u = x^2\left[\frac{1+(y/x)^3}{3+4y/x}\right]$$

Hence, z is a homogenous function of degree 2

$$z = x^n f(y/x)$$

here, $n = 2$

Now, by Euler theorem

$$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = 2z \quad \text{--- (1)} \quad \left[x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu\right]$$

here, $z = \log u$

$$\frac{\partial z}{\partial x} = \frac{1}{u}\cdot\frac{\partial u}{\partial x} \quad \text{--- (2)}$$

$$\frac{\partial z}{\partial y} = \frac{1}{u}\frac{\partial u}{\partial y} \quad \text{--- (3)}$$

putting value of eq (2) and (3) in eq (1)

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$$\frac{x}{u}\frac{\partial u}{\partial x} + \frac{y}{u}\frac{\partial u}{\partial y} = 2\log u$$

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 2u\log u$$

Proved.

Q.2. If $u = \dfrac{x^3y^3z^3}{x^3+y^3+z^3} + \log\left(\dfrac{xy+yz+zx}{x^2+y^2+z^2}\right)$,

find the value of $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} + z\dfrac{\partial u}{\partial z}$

$\Rightarrow$ Given;

$$u = \frac{x^3y^3z^3}{x^3+y^3+z^3} + \log\left(\frac{xy+yz+zx}{x^2+y^2+z^2}\right)$$

let $a = \dfrac{x^3y^3z^3}{x^3+y^3+z^2}$

$$= x^3 \cdot x^3 \left[\frac{(y/x)^3\,(z/x)^3}{1+(y/x)^3+(z/x)^3}\right]$$

$$a = x^6 \left[\frac{(y/x)^3\,(z/x)^3}{1+(y/x)^3+(z/x)^3}\right]$$

here, a is a homogeneous function of degree 6

$a = x^n f(y/x)$

$n = 6$

Akash

Page No.: _____
Date: __/__/____

by Euler theorem

$$x\frac{\partial a}{\partial x} + y\frac{\partial a}{\partial y} + z\frac{\partial a}{\partial z} = 6a \quad \text{--- (1)}$$

Now,

let $b = \log\left(\frac{xy+yz+zx}{x^2+y^2+z^2}\right)$

$$b = \log\left(\frac{y/x + y/x \cdot z/x + z/x}{1+(y/x)^2+(z/x)^2}\right)$$

here, b is ~~fx~~ homogeneous function of degree 2

$b = x^n f(y/x)$

$n = 0$

by Euler theorem,

$$x\frac{\partial b}{\partial x} + y\frac{\partial b}{\partial y} + z\frac{\partial b}{\partial z} = 0 \quad \text{--- (2)}$$

add eq-(1) and (2)

$$x\left(\frac{\partial a}{\partial x} + \frac{\partial b}{\partial x}\right) + y\left(\frac{\partial a}{\partial y} + \frac{\partial b}{\partial y}\right) + z\left(\frac{\partial a}{\partial z} + \frac{\partial b}{\partial z}\right)$$
$$= 6a$$

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 6u.$$

the value is 6a.

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i.e, $Ga = 6 \times \dfrac{x^3 y^3 z^3}{x^3+y^3+z^3}$

Q.3. If $u = \sin^{-1}\dfrac{x}{y} + \tan^{-1}\dfrac{y}{x}$, prove that

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 0$$

=) Given;

$$u = \sin^{-1}\frac{x}{y} + \tan^{-1}\frac{y}{x}$$

let $z = \sin^{-1}\dfrac{x}{y}$

$$\sin z = \frac{x}{y}$$

$$\operatorname{cosec} z = \frac{y}{x}$$

let $a = \operatorname{cosec} z = \dfrac{y}{x}$

here, a is homogeneous function of degree 0

$$a = x^n f(y/x)$$

$$n = 0$$

Akash

Page No.:
Date: / /

by euler theorem;

$$x\frac{\partial a}{\partial x} + y\frac{\partial a}{\partial y} = 0$$

$$x(-\operatorname{cosec} z \cot z)\frac{\partial z}{\partial x} + y(-\operatorname{cosec} z \cot z)\frac{\partial z}{\partial y} = 0$$

$$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial x} = 0 \quad \text{—— (1)}$$

let $\omega = \tan^{-1}\frac{y}{x}$

$$\tan\omega = \frac{y}{x}$$

let $b = \tan\omega = \frac{y}{x}$

here, b is homogeneous function of degree 0

$$b = x^n f(y/x)$$

$$n = 0$$

by euler theorem;

$$x\frac{\partial b}{\partial x} + y\frac{\partial b}{\partial y} = 0$$

$$x\sec^2\omega\frac{\partial \omega}{\partial x} + y\sec^2\omega\frac{\partial \omega}{\partial y} = 0$$

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$$x\frac{\partial w}{\partial x} + y\frac{\partial w}{\partial y} = 0 \quad \text{— (2)}$$

eq - (1) and (2), we have

$$x\left(\frac{\partial z}{\partial x} + \frac{\partial w}{\partial y}\right) + y\left(\frac{\partial z}{\partial y} + \frac{\partial w}{\partial y}\right) = 0$$

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 0$$

Hence Proved.

Q.4. If $u = \cos^{-1}\frac{x+y}{\sqrt{x}+\sqrt{y}}$, prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = -\frac{1}{2}\cot u$

$\Rightarrow$ Given;

$$u = \cos^{-1}\frac{x+y}{\sqrt{x}+\sqrt{y}}$$

$$\cos u = \frac{x+y}{\sqrt{x}+\sqrt{y}}$$

let $z = \cos u = \sqrt{x}\,\frac{(1 + y/x)}{(1+\sqrt{y/x})}$

Akash

Page No.:
Date: / /

here, z is homogeneous function of degree $1/2$

$$z = x^n f(y/x)$$

$$n = 1/2$$

by euler theorem,

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = \frac{1}{2} z$$

$$x(\sin u) \frac{\partial u}{\partial x} + y \cdot (-\sin u) \frac{\partial u}{\partial y} = \frac{1}{2} \cos u$$

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = -\frac{1}{2} \tan u \cdot \cos u$$

proved.

Q5. If $u = \tan^{-1}\left(\frac{x^2+y^2}{x-y}\right)$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{1}{2}\sin 2u$

=) Given;

$$u = \tan^{-1}\left(\frac{x^2+y^2}{x-y}\right)$$

$$\tan u = \left(\frac{x^2+y^2}{x-y}\right)$$

let $z = \tan u = (x^2+y^2/x-y)$

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$$z = \tan u = x\left(\frac{1+(y/x)^2}{1-y/x}\right)$$

here, z is a homogeneous equation of degree 1

$$z = x^n f(y/x)$$

$$n = 1$$

by euler theorem,

$$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = z$$

$$x\sec^2 u\frac{\partial u}{\partial x} + y\sec^2 u\frac{\partial u}{\partial y} = \tan u$$

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \sin u\cos u$$

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{1}{2}\sin 2u$$

Hence proved

Akash

Page No.:
Date: / /

28-2023

Q6. If $u = \sin^{-1} \dfrac{x^2+y^2}{x+y}$, prove that $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = \tan u$

Sol) Given;

$$u = \sin^{-1}\left(\frac{x^2+y^2}{x+y}\right)$$

$$\sin u = \frac{x^2+y^2}{x+y}$$

let $z = \sin u = \dfrac{x^2+y^2}{x+y}$

$$z = \sin u = x\left(\frac{1+(y/x)^2}{1+y/x}\right)$$

here, z is homogeneous function of degree 2

$$z = x^n f(y/x)$$

$$n = 1$$

by euler theorem;

$$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = z$$

$$x\cos u\frac{\partial u}{\partial x} + y\cos u\frac{\partial u}{\partial x} = \sin u$$

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial x} = \tan u$$

proved.

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